Solutions to MP 204 Problem Set 1

P.1

My own personal feeling is that, for problems that involve numbers, don't put the numbers in until you have to. In other words, be as general as possible until you have a formula for what you want to calculate, and only then put numbers in.

This problem is a good example of this, because as it turns out, we don't even need the Earth-Moon distance to get the answer: according to Newton's Law of Gravitation, the magnitude of the gravitational attractive force between them is $|\vec{F}_G| = GMm/R^2$, where M is the Earth's mass, m the moon's mass and R the distance between them. If each has a charge Q, then Coulomb's law says the force pushing them apart has magnitude $|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} Q^2/R^2$. For these to balance each other out, $|\vec{F}_G| = |\vec{F}_E|$. Note both have a $1/R^2$ in them, so it cancels out entirely:

$$\frac{GMm}{R^2} = \frac{1}{4\pi\epsilon_0}\frac{Q^2}{R^2} \Rightarrow GMm = \frac{1}{4\pi\epsilon_0}Q^2$$

so

$$Q = \sqrt{4\pi\epsilon_0 GMm}.$$

This is the formula we're after, so now we can put the numbers in:

$$Q = \sqrt{4\pi(8.85\times10^{-12}\,C^2.N^{-1}.m^{-2})(6.67\times10^{-11}\,N.m^2.kg^{-2})(5.97\times10^{24}\,kg)(7.35\times10^{22}\,kg)}$$

$$= \boxed{5.7\times10^{13}\,C}$$

which is a LOT of charge. Needless to say, there's no danger of losing the Moon due to a charge buildup any time soon...

P.2

Recall that we want $\vec{E}$ and Φ at a general point $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, not at any specific point. To do this, we only need to find the individual electric fields and electric potentials due to each charge, then add them up to get the respective field and potential for the whole configuration. To do this, remember that if a charge Q_i is located at a position $\vec{r}_i$, the electric field at a point $\vec{r}$ is

$$\vec{E}_i(\vec{r}) = \frac{Q_i}{4\pi\epsilon_0} \frac{\vec{r}-\vec{r}_i}{|\vec{r}-\vec{r}_i|^3}$$

and the potential is

$$\Phi(\vec{r}) = \frac{Q_i}{4\pi\epsilon_0} \frac{1}{|\vec{r}-\vec{r}_i|}.$$

Let's do it one charge at a time:

1: $Q_1 = q$ at $\vec{r}_1 = \vec{0}$. The electric field is

$$\vec{E}_1 = \frac{q}{4\pi\epsilon_0} \frac{(x\hat{i}+y\hat{j}+z\hat{k})-\vec{0}}{|x\hat{i}+y\hat{j}+z\hat{k}-\vec{0}|^3} = \frac{q}{4\pi\epsilon_0} \frac{x\hat{i}+y\hat{j}+z\hat{k}}{[x^2+y^2+z^2]^{3/2}}$$

and the potential is

$$\Phi_1 = \frac{q}{4\pi\epsilon_0} \frac{1}{|x\hat{i}+y\hat{j}+z\hat{k}-\vec{0}|} = \frac{q}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2+y^2+z^2}}$$

2: $Q_2 = 2q$ at $\vec{r}_2 = a\hat{i}+a\hat{j}+a\hat{k}$. The field is

$$\vec{E}_2 = \frac{2q}{4\pi\epsilon_0} \frac{(x\hat{i}+y\hat{j}+z\hat{k})-(a\hat{i}+a\hat{j}+a\hat{k})}{|(x\hat{i}+y\hat{j}+z\hat{k})-(a\hat{i}+a\hat{j}+a\hat{k})|^3}$$

$$= \frac{q}{2\pi\epsilon_0} \frac{(x-a)\hat{i}+(y-a)\hat{j}+(z-a)\hat{k}}{[(x-a)^2+(y-a)^2+(z-a)^2]^{3/2}}$$

and the potential is

$$\Phi_2 = \frac{2q}{4\pi\epsilon_0} \frac{1}{|(x\hat{i}+y\hat{j}+z\hat{k})-(a\hat{i}+a\hat{j}+a\hat{k})|} = \frac{q}{2\pi\epsilon_0} \frac{1}{\sqrt{(x-a)^2+(y-a)^2+(z-a)^2}}$$

3: $Q_3 = -3q$ at $\vec{r}_3 = -a\hat{j}+a\hat{k}$ gives

$$\vec{E}_3 = \frac{-3q}{4\pi\epsilon_0} \frac{(x\hat{i}+y\hat{j}+z\hat{k})-(-a\hat{j}+a\hat{k})}{|(x\hat{i}+y\hat{j}+z\hat{k})-(-a\hat{j}+a\hat{k})|^3}$$

$$= -\frac{3q}{4\pi\epsilon_0} \frac{x\hat{i}+(y+a)\hat{j}+(z-a)\hat{k}}{[x^2+(y+a)^2+(z-a)^2]^{3/2}}$$

and the potential is

$$\Phi_3 = \frac{-3q}{4\pi\epsilon_0} \frac{1}{|(x\hat{i}+y\hat{j}+z\hat{k})-(-a\hat{j}+a\hat{k})|} = -\frac{3q}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2+(y+a)^2+(z-a)^2}}$$

So now we can sum up the individual electric fields to get the total electric field. If we factor out a common $\frac{q}{4\pi\epsilon_0}$, we get

$$\boxed{\vec{E}(x,y,z) = \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{x}{(x^2+y^2+z^2)} + \frac{2(x-a)}{[(x-a)^2+(y-a)^2+(z-a)^2]^{3/2}} - \frac{3x}{[x^2+(y+a)^2+(z-a)^2]^{3/2}}\right]\hat{i}\right.}$$

$$+\left[\frac{y}{(x^2+y^2+z^2)^{3/2}}+\frac{2(y-a)}{[(x-a)^2+(y-a)^2+(z-a)^2]^{3/2}}+\frac{3(y+a)}{[x^2+(y+a)^2+(z-a)^2]^{3/2}}\right]\hat{j}$$

$$+\left[\frac{z}{(x^2+y^2+z^2)^{3/2}}+\frac{2(z-a)}{[(x-a)^2+(y-a)^2+(z-a)^2]^{3/2}}-\frac{3(z-a)}{[x^2+(y+a)^2+(z-a)^2]^{3/2}}\right]\hat{k}\Bigg\}$$

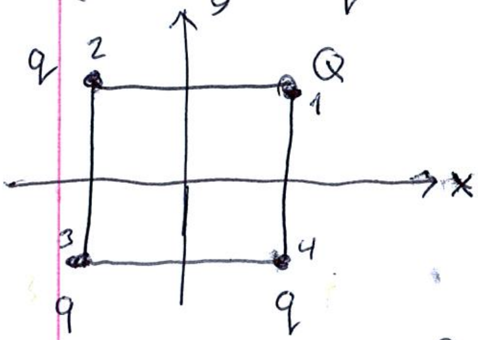
The sum of the three potentials gives a total potential of

$$\boxed{\Phi(x,y,z)=\frac{q}{4\pi\epsilon_0}\left\{\frac{1}{\sqrt{x^2+y^2+z^2}}+\frac{2}{\sqrt{(x-a)^2+(y-a)^2+(z-a)^2}}-\frac{3}{\sqrt{x^2+(y+a)^2+(z-a)^2}}\right\}}$$

(Now you can see why the potential is preferable: it's much shorter to write!)

P.3

To make things easy, I'm going to put the four charges at the vertices of a square centred on the origin in the xy-plane: in other words, its three vertices are at $\vec{r}_1=\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j}$ (where Q is located), $\vec{r}_2=-\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j}$, $\vec{r}_3=-\frac{a}{2}\hat{i}-\frac{a}{2}\hat{j}$, $\vec{r}_4=\frac{a}{2}\hat{i}-\frac{a}{2}\hat{j}$ (where the three q's are). This is shown below to the left.



All three parts of this question ask about what's going on at the origin, $\vec{r}=\vec{0}$. Thus, we can compute the total $\vec{E}$ and Φ there using the same ideas as in P.2:

Part 1: $\vec{E}_1(\vec{0})=\frac{Q}{4\pi\epsilon_0}\frac{(\vec{0}-(\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j}))}{|\vec{0}-(\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j})|^3}=-\frac{Qa}{8\pi\epsilon_0}\frac{\hat{i}+\hat{j}}{[(\frac{a}{2})^2+(\frac{a}{2})^2]^{3/2}}$

$$=\frac{Q\sqrt{2}}{4\pi\epsilon_0 a^2}(-\hat{i}-\hat{j})$$

$$\Phi_1(\vec{0})=\frac{Q}{4\pi\epsilon_0}\frac{1}{|\vec{0}-(\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j})|}=\frac{Q\sqrt{2}}{4\pi\epsilon_0 a}$$

Part 2: $\vec{E}_2(\vec{0})=\frac{q}{4\pi\epsilon_0}\frac{(\vec{0}-(-\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j}))}{|\vec{0}-(-\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j})|^3}=\frac{q\sqrt{2}}{4\pi\epsilon_0 a^2}(\hat{i}-\hat{j})$

$$\Phi_2(\vec{0})=\frac{q}{4\pi\epsilon_0}\frac{1}{|\vec{0}-(-\frac{a}{2}\hat{i}+\frac{a}{2}\hat{j})|}=\frac{q\sqrt{2}}{4\pi\epsilon_0 a}$$

Point 3: $\vec{E}_3(\vec{0}) = \frac{q}{4\pi\epsilon_0} \frac{(\vec{0} - (-\frac{a}{2}\hat{\imath} - \frac{a}{2}\hat{\jmath}))}{|\vec{0} - (-\frac{a}{2}\hat{\imath} - \frac{a}{2}\hat{\jmath})|^3} = \frac{q\sqrt{2}}{4\pi\epsilon_0 a^2}(\hat{\imath}+\hat{\jmath})$

$$\Phi_3(\vec{0}) = \frac{q}{4\pi\epsilon_0} \frac{1}{|\vec{0} - (-\frac{a}{2}\hat{\imath} - \frac{a}{2}\hat{\jmath})|} = \frac{q\sqrt{2}}{4\pi\epsilon_0 a}$$

Point 4: $\vec{E}_4(\vec{0}) = \frac{q}{4\pi\epsilon_0} \frac{(\vec{0} - (\frac{a}{2}\hat{\imath} - \frac{a}{2}\hat{\jmath}))}{|\vec{0} - (\frac{a}{2}\hat{\imath} - \frac{a}{2}\hat{\jmath})|^3} = \frac{q\sqrt{2}}{4\pi\epsilon_0 a^2}(-\hat{\imath}+\hat{\jmath})$

$$\Phi_4(\vec{0}) = \frac{q}{4\pi\epsilon_0} \frac{1}{|\vec{0} - (\frac{a}{2}\hat{\imath} - \frac{a}{2}\hat{\jmath})|} = \frac{q\sqrt{2}}{4\pi\epsilon_0 a}.$$

(Note that all three points are a distance $\frac{a}{\sqrt{2}}$ from the origin.)
Thus, the total field at the origin is

$$\vec{E}(\vec{0}) = \vec{E}_1(\vec{0}) + \vec{E}_2(\vec{0}) + \vec{E}_3(\vec{0}) + \vec{E}_4(\vec{0})$$

$$= \frac{(q - Q)\sqrt{2}}{4\pi\epsilon_0 a^2}(\hat{\imath}+\hat{\jmath})$$

($\vec{E}_2$ and $\vec{E}_4$ cancel out at the origin!) The potential at the origin is

$$\Phi(\vec{0}) = \Phi_1(\vec{0}) + \Phi_2(\vec{0}) + \Phi_3(\vec{0}) + \Phi_4(\vec{0}) = \frac{(3q+Q)\sqrt{2}}{4\pi\epsilon_0 a}.$$

Now that we have these, the problem is easy:

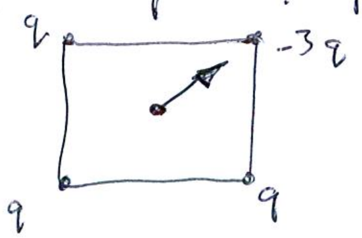
(a) For $\vec{E}(\vec{0}) = \vec{0}$, we see we simply need $\boxed{Q = q}$. So a square with equal charges at each vertex has no electric field at all in its centre.

(b) $\Phi(\vec{0}) = 0$ if $3q + Q = 0$, i.e. $\boxed{Q = -3q}$. So the last charge must be three times greater, and of opposite sign, as the other three in order to give zero potential at the centre.

(c) $Q = -3q$ for b), so if we put this into the general form for $\vec{E}(\vec{0})$ above, we obtain

$$\vec{E}(\vec{0}) = \frac{q - (-3q)}{4\pi\epsilon_0 a^2}\sqrt{2}(\hat{\imath}+\hat{\jmath}) = \boxed{\frac{q\sqrt{2}}{\pi\epsilon_0 a^2}(\hat{\imath}+\hat{\jmath})}$$

which points from point 3 toward point 1, with magnitude $\frac{2q}{\pi\epsilon_0 a^2}$.



P.4

The point of this problem is to stress the relationship between the electric field $\vec{E}$ and the electric potential Φ: given a potential, we obtain a field from it via $\vec{E}(\vec{r}) = -\vec{\nabla}\Phi(\vec{r})$; given an electric field, we get Φ from it by $\Phi(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} \vec{E}(\vec{r}')\cdot d\vec{r}'$ (where we can take the starting point $\vec{r}_0$ to be anything we like and the path from $\vec{r}_0$ to $\vec{r}$ to be any path we like).

(a) Here we're given the potential $\Phi(x,y,z) = 3x^2y\sin z$. To get $\vec{\nabla}\Phi$, we need its derivatives with respect to x, y and z:

$$\frac{\partial\Phi}{\partial x} = \frac{\partial}{\partial x}(3x^2y\sin z) = 3y\sin z\left(\frac{\partial}{\partial x}x^2\right) = 3y\sin z\,(2x)$$
$$= 6xy\sin z$$

$$\frac{\partial\Phi}{\partial y} = \frac{\partial}{\partial y}(3x^2y\sin z) = 3x^2\sin z\left(\frac{\partial}{\partial y}y\right) = 3x^2\sin z\cdot 1$$
$$= 3x^2\sin z$$

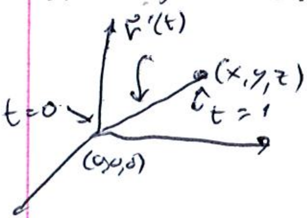
$$\frac{\partial\Phi}{\partial z} = \frac{\partial}{\partial z}(3x^2y\sin z) = 3x^2y\left(\frac{\partial}{\partial z}\sin z\right) = 3x^2y(\cos z)$$
$$= 3x^2y\cos z.$$

Thus, the electric field is

$$\vec{E}(x,y,z) = -\vec{\nabla}\Phi(x,y,z) = -\left(\frac{\partial\Phi}{\partial x}\hat{i} + \frac{\partial\Phi}{\partial y}\hat{j} + \frac{\partial\Phi}{\partial z}\hat{k}\right)$$

$$= \boxed{-6xy\sin z\,\hat{i} - 3x^2\sin z\,\hat{j} - 3x^2y\cos z\,\hat{k}}$$

(b) Now, to get the potential that gives the field $\vec{E}(x,y,z) = -x\hat{i} + y^2\hat{j} + 3z\hat{k}$.
There are loads of different ways to do the integral which will give us Φ, but here's one of the most straightforward: take $\vec{r}_0 = \vec{0}$ as the starting point, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ as the finishing point and let the path between them be a straight line. If we say the path is given by $\vec{r}'(t) = x'(t)\hat{i} + y'(t)\hat{j} + z'(t)\hat{k}$,



and starts at $t=0$ and finishes at $t=1$, then $x'(t)$ must be the line that connects 0 to x, i.e. $x'(t) = tx$. Similarly, $y'(t) = ty$ and $z'(t) = tz$.

Therefore, on this path,

$$\vec{E}(\vec{r}'(t)) = -x'(t)\hat{i} + (y'(t))^2\hat{j} + 3z'(t)\hat{k}$$
$$= -xt\hat{i} + y^2t^2\hat{j} + 3zt\hat{k}$$

The line element on the path is

$$d\vec{r}'(t) = \left(\frac{dx'(t)}{dt}\hat{i} + \frac{dy'(t)}{dt}\hat{j} + \frac{dz'(t)}{dt}\hat{k}\right)dt = (x\hat{i} + y\hat{j} + z\hat{k})\,dt$$

Therefore,

$$\vec{E}(\vec{r}'(t))\, d\vec{r}'(t) = (-xt\,\hat{i} + y^2t^2\,\hat{j} + 3zt\,\hat{k}) \cdot (x\,\hat{i} + y\,\hat{j} + z\,\hat{k})\, dt$$
$$= (-x^2t + y^3t^2 + 3z^2t)\, dt.$$

Thus, we just integrate this for $t=0$ and $t=1$ and multiply by -1 to get Φ:

$$\Phi(x,y,z) = -\int_{\vec{r}_0}^{\vec{r}} \vec{E}(\vec{r}')\cdot d\vec{r}' = -\int_0^1 (-x^2t + y^3t^2 + 3z^2t)\, dt$$

$$= -\left(-\frac{1}{2}x^2t^2 + \frac{1}{3}y^3t^3 + \frac{3}{2}z^2t^2\right|_0^1$$

$$= \boxed{\frac{1}{2}x^2 - \frac{1}{3}y^3 - \frac{3}{2}z^2}$$

and you can then confirm that this agrees with $\vec{E} = -\vec{\nabla}\Phi$ (always a good idea to check this!)